Anshuman Pandey

AI ENGINEER / DATA SCIENTIST

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PROFESSIONAL SUMMARY

CSE (AI/ML) student at VIT Bhopal with a strong foundation in deep learning, computer vision, reinforcement learning, and LLM-based systems. Built a chess engine from scratch using MCTS and reinforcement learning, a retinal disease classifier fine-tuned to 98.2% accuracy, and a retrieval-augmented generation (RAG) pipeline for document question-answering. Comfortable across the applied-AI workflow — model design, evaluation, and deployment via FastAPI and Gradio — with a statistics-driven approach to experimentation. Proficient in Python, PyTorch, TensorFlow, and LLM/RAG tooling (LangChain, FAISS). Actively seeking internship or entry-level roles in AI Engineering or Data Science.

EXPERIENCE

ServiceNow Virtual Intern | *May 2026*

- Completed a virtual internship focused on ServiceNow platform administration and automation, gaining practical exposure to Flow Designer, Automated Test Framework (ATF), Reporting, and Agentic AI concepts.

PROJECTS

RAG Document Q&A System | Python, LangChain, FAISS, Gemini, Gradio, MLflow | 2026

- Built a retrieval-augmented generation pipeline for document question-answering, using PyMuPDF for parsing and FAISS with MMR retrieval for diverse, relevant context selection.
- Improved answer relevance with a cross-encoder re-ranking stage ahead of a Gemini LLM for generation, exposed through a Gradio UI backed by a FastAPI service.
- Tracked experiments with MLflow and automated testing via GitHub Actions CI.
  https://github.com/AnshumanJ28/rag-document-qa

RAG Document Intelligence Chatbot | Python, LangChain, FAISS, Groq (Llama 3.1), MLflow, Gradio | 2026

- Built a second, production-style RAG pipeline with a two-stage retrieval design: MMR fetches a diverse top-10 candidate set, then a cross-encoder re-ranker narrows it to the top 4 for precision.
- Implemented a hallucination guard using rerank-score thresholds — off-topic queries score sharply lower (e.g. -11 vs. +7 for on-topic questions), and the model correctly declines to answer when nothing relevant is retrieved.
- Logged latency, rerank scores, and answer length per query with MLflow to compare configurations (chunk size, top-k, LLM model) systematically.
  https://github.com/AnshumanJ28/rag-document-chatbot

Recommendation System & A/B Testing Platform | Python, ALS, Streamlit, FastAPI, Cloudflare Tunnel | 2026

- Built an ALS-based recommender system paired with a statistically rigorous A/B testing framework, using MD5 hash-based user assignment for consistent, reproducible experiment bucketing.
- Implemented z-tests and power analysis to validate experiment significance and sizing, exposing results through an interactive Streamlit dashboard.
- Served the recommendation model via a FastAPI backend and deployed the full system with Cloudflare Tunnel for public access.
  https://github.com/AnshumanJ28/recsys-ab-testing

TECHNICAL SKILLS

ML / AI: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, CNN, LSTM, FFT preprocessing
LLM & RAG: LangChain, FAISS, Cross-Encoder Re-ranking, Gemini API, Prompt Engineering
Languages: Python, C, C++, Java, SQL (MySQL), HTML, CSS
MLOps & Deployment: FastAPI, Docker, MLflow, GitHub Actions (CI/CD), Gradio
Concepts: Data Structures & Algorithms, Computer Vision, Reinforcement Learning

EDUCATION

B.Tech in Computer Science & Engineering (AI/ML) — Vellore Institute of Technology, Bhopal
2023 – Present | CGPA: 8.52/10 • Class XII: 89.5% • Class X: 89.4%

CERTIFICATIONS

C & C++ Programming (Udemy, HackerRank)|**Python & Java** (HackerRank)|**Cloud Computing** (NPTEL)|**Advanced Excel** (Udemy) |**AI & Agentic AI** (TalentSprint)|**ML** (Coursera)|**AWS Trainer & AWS Academy** (AWS)|**IoT** (NPTEL)|**ServiceNow**

ACHIEVEMENTS

- **2nd Runner-up**, IEEE-NSUT Ideathon — awarded for ML-based project among 80+ competing teams.
- **Top 10 finish**, BITS Pilani Hackathon — among top teams nationally in a competitive problem-solving event.